

The sharp integrability range for trace-normalized parabolic operators

Research proof

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Abstract

The exact admissible range is $(\max\{(n+2)/2, n[1 - (n-1)\delta]\}, \infty)$. The upper bound follows from a sharp matrix inequality modulo positive Hessian boundary fluxes, integrated over shifted $(n-1)$ -Hessian Dirichlet resolvents. Convex source decomposition supplies the required Mellin-energy estimates. A ball inverse yields the maximum estimate by expanding-annuli localization and an interior approximation to the precise measurable coefficient class. Explicit normalized constructions exclude every smaller exponent and both possible critical endpoints.

1 Exact statement and proof structure

Fix an integer $n \geq 2$ and $0 < \delta < 1/n$. Let

$$B_1 = \{x \in \mathbb{R}^n : |x| < 1\}, \quad Q_1 = B_1 \times (0, 1),$$

with Euclidean norm, and let

$$\partial_{\text{par}} Q_1 = (\overline{B_1} \times \{0\}) \cup (\partial B_1 \times [0, 1]).$$

Throughout, the coefficient class is exactly all Lebesgue-measurable symmetric fields $A = (a_{ij})$ on Q_1 such that almost everywhere

$$\text{tr } A = 1, \quad \xi^T A \xi \geq \delta |\xi|^2 \quad \text{for every } \xi \in \mathbb{R}^n. \quad (1)$$

Set

$$\mathcal{P}_A u = u_t - \sum_{i,j} a_{ij} u_{ij}, \quad \Lambda = 1 - (n-1)\delta, \quad N = n+2, \quad p_c = N/2.$$

Thus every allowed A also satisfies $A \leq \Lambda I$. A finite $p \geq 1$ is admissible if a constant depending only on n, δ, p bounds

$$\max_{\overline{Q_1}} |u| \leq C \|\mathcal{P}_A u\|_{L^p(Q_1)} \quad (2)$$

for every such A and every real-valued

$$u \in C(\overline{Q_1}) \cap C^{2,1}(Q_1), \quad u = 0 \text{ on } \partial_{\text{par}} Q_1, \quad \mathcal{P}_A u \in L^p(Q_1).$$

Here $C^{2,1}$ means twice continuously differentiable spatially and once continuously differentiable in time; no boundary derivative bound or additional Sobolev assumption is imposed.

Theorem 1.1 (Sharp range). *With precisely these hypotheses,*

$$\mathcal{E}_{\text{par}}(n, \delta) = \left(\max \left\{ \frac{n+2}{2}, n[1 - (n-1)\delta] \right\}, \infty \right).$$

For every excluded $p \in [1, \infty)$ there are smooth admissible matrices A_j and functions satisfying the stated boundary and regularity conditions such that

$$\max_{\overline{Q_1}} |u_j| = 1, \quad \left\| \mathcal{P}_{A_j} u_j \right\|_{L^p(Q_1)} \longrightarrow 0.$$

In particular the infimum is the displayed maximum and is never admissible.

The upper argument will first establish a zero-boundary ball inverse at any auxiliary exponent

$$2 < r < n, \quad r > n\Lambda \quad (n \geq 3). \quad (3)$$

Dimension two has a direct $r = 2$ inverse. A separate elementary localization then requires only $p \geq r$ and $p > p_c$. Finally an interior approximation transfers the estimate to exactly (1) and the test class above.

2 Elliptic inputs and shifted Hessian energies

Assume for now $n \geq 3$ and put $m = n - 1$. For a symmetric matrix H , let $\sigma_j(H)$ be the unnormalized j th elementary symmetric polynomial, $\sigma_0 = 1$, and put

$$P_j(H) = \frac{\sigma_j(H)}{\binom{n}{j}}.$$

Write Γ_j for the Gårding cone containing the positive definite matrices, equivalently the component on which $\sigma_1, \dots, \sigma_j$ are positive. We use its standard convexity, ellipticity, Maclaurin inequalities, and Newton-tensor identities. The two analytic elliptic inputs are the following classical theorems, in these precise forms:

1. On a ball, a smooth strictly positive right side has a smooth zero-Dirichlet j -Hessian solution in Γ_j , unique in that branch. The linearization at each fixed such solution is strictly elliptic. This is the classical Caffarelli–Nirenberg–Spruck Dirichlet theorem [1]; balls satisfy its boundary curvature hypotheses.
2. For a zero-boundary j -admissible potential on B_R , with $\mathcal{H}_j(u) = \int (-u) P_j(D^2 u)$,

$$\|u\|_{L^{j+1}(B_R)} \leq C(n, j) R^{2j/(j+1)} \mathcal{H}_j(u)^{1/(j+1)}. \quad (4)$$

This is the indicated exponent of the Hessian Sobolev inequality [2]. The theorem permits this exponent for every $1 \leq j \leq n$; for $j > n/2$ it even permits the supremum norm. No endpoint beyond (4) is used here.

All later constants are independent of the elliptic solution's derivative bounds. Those bounds are used only to justify identities for fixed smooth data.

Fix a ball $D = B_R$, a smooth nonnegative w on its closure, and $\tau > 0$. Let $u_\tau = 0$ on ∂D solve

$$P_m(D^2 u_\tau + \tau I) - \tau^m = w, \quad D^2 u_\tau + \tau I \in \Gamma_m. \quad (5)$$

Indeed $v = u_\tau + \tau(|x|^2 - R^2)/2$ is the usual m -Hessian potential with right side $w + \tau^m > 0$. Thus even when w vanishes, every fixed positive τ is nondegenerate. Maclaurin gives

$$\Delta u_\tau \geq n[(w + \tau^m)^{1/m} - \tau] \geq 0, \quad u_\tau \leq 0, \quad \partial_\nu u_\tau \geq 0. \quad (6)$$

If u_0 is the unshifted potential for positive data, comparison gives $u_0 \leq u_\tau \leq 0$.

The polynomial in (5) is

$$L_\tau(H) = \sum_{j=1}^m \binom{m}{j} \tau^{m-j} P_j(H).$$

Define on the convex shifted-admissible zero-boundary class

$$J_\tau(u) = \sum_{j=1}^m \frac{\binom{m}{j}}{j+1} \tau^{m-j} \int_D (-u) P_j(D^2 u).$$

The divergence-free Newton tensors and zero-boundary variations give

$$DJ_\tau(u)[h] = - \int_D h L_\tau(D^2 u), \quad (7)$$

$$D^2 J_\tau(u)[h, h] = \int_D DP_m(D^2 u + \tau I)[Dh, Dh] > 0 \quad (h \neq 0). \quad (8)$$

Thus the solution maximizes the concave functional $\int (-u)w - J_\tau(u)$. Its value

$$F_\tau(w) = \int_D (-u_\tau)w - J_\tau(u_\tau) \quad (9)$$

is nonnegative and convex in w , has $F_\tau(0) = 0$, and satisfies

$$DF_\tau(w)[h] = \int_D (-u_\tau)h. \quad (10)$$

These follow also by differentiation, since (7) cancels the variation through u_τ . Smooth dependence holds locally for each fixed positive τ by the elliptic linearization. Integrating (10) along a positive amplitude ray and using comparison gives

$$0 \leq F_\tau(w) \leq F_0(w) = \frac{m}{m+1} \mathcal{H}_m(u_0) \leq C_n R^2 \|w\|_{L^{(m+1)/m}}^{(m+1)/m}. \quad (11)$$

The final inequality follows from (4) and Hölder; the nonnegative-data version follows by approximation from $w + \varepsilon$.

Lemma 2.1 (Quadratic bound). *Every shifted-admissible zero-boundary potential obeys*

$$J_\tau(u) \geq \frac{m}{n(m+1)} \tau^{m-1} \|Du\|_{L^2}^2.$$

Consequently

$$F_\tau(w) \leq \frac{n(m+1)}{4m} \tau^{1-m} \|w\|_{H^{-1}(D)}^2 \leq C_n R^2 \tau^{1-m} \|w\|_{L^2(D)}^2. \quad (12)$$

Proof. Put $K = D^2 u + \tau I$. Along the segment from zero to u , its shifted Hessian is $K_t = (1-t)\tau I + tK$. For each unit e , the derivative of P_m in the rank-one direction $e \otimes e$ is the normalized σ_{m-1} of the

compression to $e^\perp$. Its shifted polynomial has nonnegative coefficients because $K \in \Gamma_m$. Retaining only its constant term gives

$$DP_m(K_t)[e, e] \geq \frac{\binom{n-1}{m-1}}{\binom{n}{m}}(1-t)^{m-1}\tau^{m-1} = \frac{m}{n}(1-t)^{m-1}\tau^{m-1}.$$

Equivalently this follows from positivity of all lower Newton tensors on Γ_m . Since $J_\tau(0) = DJ_\tau(0) = 0$, Taylor's integral formula and (8) yield the factor $(m/n) \int_0^1 (1-t)^m dt = m/[n(m+1)]$. Taking the supremum of the linear-minus-quadratic expression over all $H_0^1(D)$ gives the first bound in (12) by completing the Hilbert-space square. Poincaré gives the second. $\square$

3 Mellin source estimates in the full real range

Fix

$$0 < s < m-1, \quad \alpha = \frac{1+s}{m}, \quad q = 1 + \alpha, \quad r = \frac{q}{\alpha} = \frac{n+s}{1+s}. \quad (13)$$

Thus $2 < r < n$ and $q = r/(r-1)$. Put

$$F_s(w) = \int_0^\infty \tau^{s-1} F_\tau(w) d\tau, \quad \Psi_s(w) = - \int_0^\infty \tau^{s-1} u_\tau(w) d\tau.$$

Lemma 3.1 (Energy and potential mapping). *For smooth nonnegative data,*

$$F_s(w) \leq C_{n,s} R^2 \|w\|_{L^q}^q, \quad \|\Psi_s(w)\|_{L^r} \leq C_{n,s} R^2 \|w\|_{L^q}^{q-1}. \quad (14)$$

Both quantities are nonnegative, and Ψ_s vanishes on the boundary.

Proof. Choose a smooth $\theta : [0, \infty) \rightarrow [0, 1]$, equal to one on $[0, 1]$ and zero on $[2, \infty)$. For each τ split

$$w_\ell = w\theta(w/\tau^m), \quad w_h = w - w_\ell.$$

Convexity, (11), and (12) give, with $q_0 = (m+1)/m$,

$$F_\tau(w) \leq \frac{1}{2} F_\tau(2w_\ell) + \frac{1}{2} F_\tau(2w_h) \leq C_n R^2 \left[\tau^{1-m} \int_{w \leq 2\tau^m} w^2 + \int_{w > \tau^m} w^{q_0} \right].$$

For a scalar $a > 0$, Tonelli reduces the two Mellin integrals to

$$a^{q_0} \int_0^{a^{1/m}} \tau^{s-1} d\tau = \frac{a^q}{s}, \quad a^2 \int_{(a/2)^{1/m}}^\infty \tau^{s-m} d\tau = \frac{2^{(m-1-s)/m}}{m-1-s} a^q.$$

This proves the first estimate in every dimension; no reversed interpolation between endpoint norms is used.

For bounded data $0 \leq w \leq M$, comparison with the quadratic having Hessian cI , where $c = (\tau^m + M)^{1/m} - \tau$, gives

$$0 \leq -u_\tau(x) \leq \frac{1}{2}(R^2 - |x|^2) \min \left\{ M^{1/m}, \frac{M}{m\tau^{m-1}} \right\}. \quad (15)$$

The Mellin integral converges uniformly, has zero boundary values, and is bounded by $C_{n,s}(R^2 - |x|^2)M^{q-1}$. For a finite parameter interval $[\varepsilon, L]$, differentiating the convex energy gives its potential

first variation. The positive directional convexity inequality then gives, for $h \geq 0$ and $\lambda = \|w\|_{L^q} / \|h\|_{L^q}$,

$$\int_D \Psi_{s;\varepsilon,L}(w)h \leq \frac{F_{s;\varepsilon,L}(w + \lambda h) - F_{s;\varepsilon,L}(w)}{\lambda} \leq 2^q C_{n,s} R^2 \|w\|_{L^q}^{q-1} \|h\|_{L^q}.$$

The zero-data case is immediate. Monotone convergence in the positive potential and positive L^r duality prove the second estimate in (14). This finite-parameter argument does not require uniform Hessian convergence at $\tau = 0$ when w vanishes. $\square$

4 The sharp matrix envelope and its Mellin integral

For a symmetric H define the null-minor polynomial

$$\mathcal{N}_\tau(H) = \frac{\det(H + \tau I) - \tau^{n-1} \operatorname{tr} H - \tau^n}{n} = \frac{1}{n} \sum_{j=2}^n \tau^{n-j} \sigma_j(H). \quad (16)$$

For $X \geq 0$ put

$$T_\tau(X) = (\tau^m + X)^{1/m} - \tau, \quad R_\tau(X) = \frac{(\tau + T_\tau(X))^n - \tau^n - n\tau^m T_\tau(X)}{n} \geq 0.$$

Lemma 4.1 (Pointwise sharp envelope). *Suppose $K = H + \tau I \in \Gamma_m$ and $P_m(K) = w + \tau^m$ with $w \geq 0$. For every matrix in (1),*

$$wA : H - \mathcal{N}_\tau(H) \geq wT_\tau(n\Lambda w) - R_\tau(n\Lambda w). \quad (17)$$

Proof. Let ℓ be the least eigenvalue of K , and write its other eigenvalues as $y_1, \dots, y_m$. They are strictly positive. One direct justification is to consider $Q(t) = \det(tI + K)$. Because $\sigma_1(K), \dots, \sigma_{n-1}(K) > 0$, its derivative is positive for $t \geq 0$. Two nonpositive eigenvalues of K would give two nonnegative roots of Q , counting multiplicity, and hence a nonnegative root of Q' by Rolle's theorem, a contradiction. Put

$$S = \sum_i y_i, \quad P = \prod_i y_i, \quad C = \sigma_m(K) = n(w + \tau^m).$$

The minimum of $A : H$ is obtained by putting Λ in the least eigendirection and δ in the others. Apart from constants depending only on w, τ , the left side of (17) is therefore

$$aS + (w\Lambda + \tau^m/n)\ell - \ell P/n, \quad a = w\delta + \tau^m/n > 0, \quad (18)$$

under $C = P + \ell\sigma_{m-1}(y)$.

Fix ℓ and let $v > 0$ be the positive-value root of

$$v^m + m\ell v^{m-1} = C. \quad (19)$$

For $\ell < 0$ this root lies above $-m\ell$, where the left side is strictly increasing; for $\ell \geq 0$ it is increasing for all positive v . Also $v \geq \ell$: if $\ell > 0$, the original ordering gives $C \geq n\ell^m$, and otherwise this is immediate.

If $\ell \leq 0$, arithmetic-geometric mean applied to $1/y_i$ gives $C \leq P + m\ell P^{(m-1)/m}$, hence $P \geq v^m$ and $S \geq mv$. Both aS and $-\ell P/n$ in (18) are then minimized by equalizing the y_i to v . If $\ell \geq 0$, the same inequality gives $P \leq v^m$, whereas Maclaurin at fixed sum gives

$$C \leq (S/m)^m + m\ell(S/m)^{m-1},$$

so again $S \geq mv$, and the same conclusion follows. The replacement $(\ell, v, \dots, v)$ remains in Γ_m : in the negative case (19) gives $\ell > -v/m$, which is its strongest required elementary-symmetric inequality. Thus the full matrix minimum reduces rigorously to equal upper eigenvalues.

Write $W = w + \tau^m$ and parameterize this curve by

$$\ell(v) = \frac{nW - v^m}{mv^{m-1}}, \quad \ell'(v) = -\frac{1 + (m-1)nW/v^m}{m} < 0, \quad X = v^m - \tau^m.$$

Its admissible range is $X \geq w$, equivalent to $\ell \leq v$. For the resulting scalar remainder $G(v) = wA : H - \mathcal{N}_\tau(H)$, with the minimizing A , differentiation gives

$$G'(v) = (w\Lambda - X/n)(\ell'(v) - 1). \quad (20)$$

For verification, its ℓ coefficient is $w\Lambda - X/n$; the remaining derivative is exactly its negative, leaving (20). Since $\ell' - 1 < 0$, the unique minimum is $X = n\Lambda w$, which is allowed because $n\Lambda > 1$. Substituting there yields

$$[w(1 - \Lambda) + (m/n)\tau^m](v - \tau) - \tau\Lambda w = wT_\tau(n\Lambda w) - R_\tau(n\Lambda w).$$

This proves the claim. When $w = 0$ the same minimum is at $H = 0$ and has value zero. $\square$

Lemma 4.2 (Exact scalar Mellin cancellation). *With (13), put*

$$C_{m,s} = \frac{B(s/m, (m-1-s)/m)}{m(s+1)} > 0.$$

Then

$$\int_0^\infty \tau^{s-1} T_\tau(X) d\tau = C_{m,s} X^\alpha, \quad (21)$$

$$\int_0^\infty \tau^{s-1} R_\tau(X) d\tau = C_{m,s} \frac{\alpha}{q} X^q. \quad (22)$$

Consequently the integral of the right side of (17) is

$$c_* w^q, \quad c_* = C_{m,s} (n\Lambda)^\alpha (1 - n\Lambda/r). \quad (23)$$

It is positive exactly when $r > n\Lambda$.

Proof. Differentiate the first positive integral with respect to X and substitute $\tau = X^{1/m} t$. The beta integral is

$$\frac{1}{m^2} B(s/m, (m-1-s)/m) X^{\alpha-1}.$$

Integration from zero proves (21). All endpoint integrals converge because $0 < s < m-1$. Direct differentiation gives $\partial_X R_\tau(X) = X \partial_X T_\tau(X)$ and $R_\tau(0) = 0$. Tonelli, with nonnegative integrands, therefore proves (22). Using $X = n\Lambda w$ gives $C_{m,s} X^\alpha [w - (\alpha/q)X]$, which is (23). In particular no separately divergent scalar terms have been subtracted. The absolute integral of the envelope is bounded by the sum of the two positive integrals and hence by Cw^q . $\square$

For any smooth zero-boundary u on a ball with $a = \partial_\nu u \geq 0$, Newton's divergence identity gives

$$\int_{B_R} \sigma_j(D^2 u) = \frac{\binom{n-1}{j-1}}{j R^{j-1}} \int_{\partial B_R} a^j \geq 0. \quad (24)$$

Indeed $j\sigma_j = \partial_i(\sigma_j^{ij} u_j)$, the gradient is $a\nu$ on the sphere, and the tangential Hessian is $(a/R)I$. This proves the formula regardless of the interior sign of the minor. In particular (6) and (16) show that all the null-minor terms in Lemma 4.1 have favorable integrated sign for $u = u_\tau$.

5 The strong ball inverse

Proposition 5.1. *For $n \geq 3$ and every r in (3), every smooth coefficient field satisfying (1) has the estimate*

$$\|v\|_{L^r(B_R \times (a,b))} \leq K(n, \delta, r) R^2 \|\mathcal{P}_A v\|_{L^r(B_R \times (a,b))} \quad (25)$$

for its zero-initial, zero-lateral solution. The constant is independent of R , $b - a$, and coefficient derivatives. In dimension two the same assertion holds at $r = 2$.

Proof in dimensions $n \geq 3$. Choose $s = (n - r)/(r - 1)$, so (13) holds and $c_* > 0$. First consider the forward killed adjoint equation, with smooth nonnegative source f , zero spatial boundary values and zero initial data:

$$w_t = \partial_i \partial_j (a_{ij} w) + f.$$

Classical smooth linear theory gives a nonnegative solution, smooth up to the boundary for compatible data. It suffices initially to take f smooth, spatially supported away from the boundary, and zero near initial time. For each positive τ its shifted potential is nondegenerate even where $w = 0$.

Fix $0 < \varepsilon < L < \infty$ and write

$$E_{\varepsilon, L}(w) = \int_{\varepsilon}^L \tau^{s-1} F_{\tau}(w) d\tau, \quad \Psi_{\varepsilon, L}(w) = - \int_{\varepsilon}^L \tau^{s-1} u_{\tau}(w) d\tau.$$

The classical finite-parameter chain rule and (10) give

$$\frac{d}{dt} E_{\varepsilon, L}(w) + \int_{\varepsilon}^L \tau^{s-1} \int_D w A : D^2 u_{\tau} dx d\tau = \int_D \Psi_{\varepsilon, L}(w) f.$$

There is no boundary term: both u_{τ} and the original density w vanish on the spatial boundary. Apply Lemma 4.1 and (24), then integrate in time. The final energy is nonnegative and the initial energy is zero. Let $\varepsilon \downarrow 0$ and $L \uparrow \infty$. The energies and nonnegative forcing terms pass by monotone convergence. The explicit scalar envelope passes by dominated convergence: Lemma 4.2 bounds its absolute parameter integral by Cw^q , and this fixed smooth evolution is bounded on a finite cylinder. It follows that

$$F_s(w(b)) + c_* \int_a^b \|w(t)\|_{L^q(D)}^q dt \leq \int_a^b \int_D \Psi_s(w) f. \quad (26)$$

No limiting Hessian at $\tau = 0$ and no regularization $w + \eta$ were used. By Lemma 3.1, Hölder in space and time, and $1/q + 1/r = 1$, the right side is at most

$$C_{n,s} R^2 \|w\|_{L^q(D \times (a,b))}^{q-1} \|f\|_{L^q(D \times (a,b))}.$$

After discarding the final energy and dividing when the density norm is nonzero, this gives

$$\|w\|_{L^q(D \times (a,b))} \leq C(n, \delta, r) R^2 \|f\|_{L^q(D \times (a,b))}.$$

Positivity extends this estimate to signed data by domination and approximation. Time reversal followed by ordinary adjoint duality gives (25). The coefficient cone is unchanged by time reversal. All constants in the resulting estimate depend only on the displayed parameters. Smoothness bounds were used only for each fixed equation and finite-parameter identity, not in its inequalities. $\square$

Proof in dimension two. For each nonnegative adjoint density w , solve $\Delta u = w$ on B_R with $u = 0$ on the boundary. Let $\lambda_1 \leq \lambda_2$ be its Hessian eigenvalues. Since $w = \lambda_1 + \lambda_2 \geq 0$,

$$wA : D^2u \geq \delta w^2 + (1 - 2\delta)w\lambda_1 = \delta w^2 + (1 - 2\delta)(\det D^2u + \lambda_1^2).$$

Formula (24) gives a nonnegative integral of the determinant, so $\int wA : D^2u \geq \delta \int w^2$. The dual Poisson energy $F(w) = \frac{1}{2} \int |Du|^2$ has first variation $-u$, and Poincaré gives $\|u\|_{L^2} \leq CR^2 \|w\|_{L^2}$. The same adjoint integration by parts therefore yields the zero-initial L^2 inverse. Here the Poisson problem is nondegenerate without any parameter or source regularization. Duality proves the asserted primal estimate at $r = 2$. $\square$

6 Local homogeneous control from the contact estimate

We record the elementary parabolic ABP form used both here and in the final approximation. On $B_R \times (a, b)$, a smooth v satisfies

$$\sup v \leq \sup_{\partial_{\text{par}}} v + C(n, \delta) R^{n/(n+1)} \|(\mathcal{P}_A v)_+\|_{L^{n+1}}. \quad (27)$$

Its constant does not depend on $b - a$. Moreover, when the parabolic boundary values are nonpositive, the integral can be restricted to the positive global parabolic upper contact set. For completeness, subtract the boundary supremum and put $M = \sup v > 0$. For every $|\xi| < M/(4R)$ and $h \in (M/4, M/2)$, the first time at which $\max_x (v(x, t) - \xi \cdot x)$ reaches h gives an interior contact. The map

$$(x, t) \mapsto (Dv, v - x \cdot Dv)$$

therefore covers a set of volume $|B_1| M^{n+1} / (4^{n+1} R^n)$. On these contacts $v_t \geq 0$, $H = -D^2v \geq 0$, and its absolute Jacobian is $v_t \det H$. The n eigenvalues of $A^{1/2} H A^{1/2}$ together with v_t have sum $\mathcal{P}_A v$. Arithmetic–geometric mean and $\det A \geq \delta^n$ give

$$v_t \det H \leq \frac{(\mathcal{P}_A v)_+^{n+1}}{(n+1)^{n+1} \delta^n}.$$

The area formula proves (27), with the contact restriction. The calculation can first be performed for fully smooth v . Local mollification converging uniformly with v_t, D^2v extends the unrestricted estimate to $C^{2,1}$ functions on compact cylinders, also for measurable bounded A .

Lemma 6.1 (Homogeneous local estimate). *For $s \geq 2$, a smooth solution of $\mathcal{P}_A z = 0$ on the backward cylinder $C_r(x_0, t_0) = B_r(x_0) \times (t_0 - r^2, t_0]$ satisfies*

$$|z(x_0, t_0)| \leq C(n, \delta, s) r^{-N/s} \|z\|_{L^s(C_r(x_0, t_0))}. \quad (28)$$

Proof. By scaling it suffices to work on $B_1 \times [-1, 0]$. Put $m = n + 1$, $\zeta = (1 - |x|^2)(t + 1)$, $\eta = \zeta^m$, and $v = \eta z$. It vanishes on the parabolic boundary. At a positive global upper contact of v , a spatial supporting affine plane dominates all earlier values. In particular it is nonnegative on the spatial sphere, since $v = 0$ there. Consequently

$$|Dv| \leq \frac{v}{1 - |x|}, \quad |Dz| \leq C_m \frac{z}{1 - |x|}$$

at such a contact, where $z > 0$. Product differentiation gives $\mathcal{P}_A v = z \mathcal{P}_A \eta - 2AD\eta \cdot Dz$. Direct differentiation of the cutoff gives

$$|\mathcal{P}_A \eta| + C_m \frac{|D\eta|}{1 - |x|} \leq C_m \eta^{1-2/m}.$$

For the time term use $\eta^{2/m}/(t+1) = (1-|x|^2)^2(t+1) \leq 1$; for spatial terms use $\eta^{2/m}/(1-|x|)^2 \leq 4$. Thus on these contacts

$$(\mathcal{P}_A v)_+ \leq C_m z \eta^{1-2/m} = C_m v^{1-2/m} z^{2/m}.$$

Applying contact-set ABP, with $M = \sup v_+$, gives

$$M \leq C M^{1-2/m} \left(\int (z_+)^{2(n+1)/m} \right)^{1/(n+1)}.$$

If $M > 0$, divide and use $m = n + 1$ to obtain $M \leq C \|z_+\|_{L^2}$. The cutoff is bounded below on $B_{1/2} \times [-1/4, 0]$. Apply the same argument to $-z$. This proves the L^2 version, including the central terminal point by continuity. Hölder on C_r , whose volume is $|B_1| r^N$, then gives (28) for every $s \geq 2$. $\square$

7 The expanding-annuli localization

Proposition 7.1 (Ball inverse implies maximum control). *Suppose $s \geq 2$ and every smooth admissible coefficient has the zero-boundary estimate*

$$\|z\|_{L^s(B_R \times I)} \leq K R^2 \|\mathcal{P}_A z\|_{L^s(B_R \times I)} \quad (29)$$

on all balls and all finite time intervals, with K independent of radius, time length, and coefficient derivatives. Then for every finite $p \geq s$ with $p > p_c$, the bounded causal whole-space potential of a smooth source supported in $B_1 \times (0, 1)$ satisfies

$$\|v\|_\infty \leq C(n, \delta, s, p, K) \|g\|_{L^p}. \quad (30)$$

Here coefficients are globally smooth with bounded derivatives.

Proof. First let $g \geq 0$, and extend it by zero before time zero. The bounded causal solution exists for each fixed smooth coefficient by the classical parabolic theory, equivalently the time-reversed diffusion representation. For evaluation at (x, t) the diffusion starts at x and has generator $A(\cdot, t - \tau) : D^2$ at elapsed time τ . It has no drift; bounded diffusion coefficients prevent explosion. Positivity and the bounded-source comparison give

$$0 \leq v \leq \|g\|_\infty, \quad V := \sup_{x \in \mathbb{R}^n, t \leq 1} v(x, t) < \infty.$$

By stopping at first entrance into $\overline{B_1}$, where all source is supported, this supremum equals the supremum over $\overline{B_1} \times [0, 1]$. Indeed a path starting outside collects no forcing before entrance; its remaining potential at entrance is bounded by that inside supremum. No maximum need be attained.

Fix (x_0, t_0) in this latter set. Let $r_j = 4 \cdot 2^{-j}$. Choose smooth nonincreasing $\chi, \psi : [0, \infty) \rightarrow [0, 1]$ such that $\chi = 1$ on $[0, 1]$, $\chi = 0$ on $[2, \infty)$, $\psi = 1$ on $[0, 1]$, and $\psi = 0$ on $[3, \infty)$. Take χ constant near zero, so the radial spatial cutoff is smooth. For $t \leq t_0$ put

$$q_j(x, t) = \chi\left(\frac{|x - x_0|}{r_j}\right) \psi\left(\frac{t_0 - t}{r_j^2}\right), \quad g_j = (q_j - q_{j+1})g.$$

Then $q_0 g = g$ at all relevant past times and $0 \leq g_j \leq g$. The support of g_j has measure at most $C_n r_j^N$, lies in

$$B_{2r_j}(x_0) \times (t_0 - 3r_j^2, t_0],$$

and avoids $C_{r_j/2}(x_0, t_0)$. The finite identity $g = \sum_{j=0}^{M-1} g_j + q_M g$ is exact in the past of t_0 . Its corresponding causal potential identity is justified by linearity and bounded-solution uniqueness. The remainder obeys

$$0 \leq v_{\text{rem}}(x_0, t_0) \leq 3r_M^2 \|g\|_\infty \longrightarrow 0, \quad (31)$$

before any desired L^p bound is used.

Write $\alpha = 2 - N/p > 0$, choose $0 < \beta < \alpha/2$, and choose $K_1 \geq 4$ large enough that

$$\frac{8K_1^{-2}}{1 - 2^{-2\beta}} \leq \frac{1}{2}.$$

Set $R_j = K_1 2^{\beta j} r_j$ and let z_j be the solution killed on $B_{R_j}(x_0)$, with source g_j and zero initial value at $t_0 - 4r_j^2$. The source is supported away from the spatial boundary and vanishes near the strip bottom. The solution is homogeneous on $C_{r_j/2}$. Lemma 6.1, (29), and Hölder on the source support yield

$$\begin{aligned} z_j(x_0, t_0) &\leq C r_j^{-N/s} K R_j^2 \|g_j\|_{L^s} \\ &\leq C K K_1^2 2^{2\beta j} r_j^{2-N/p} \|g_j\|_{L^p}. \end{aligned} \quad (32)$$

The condition $p \geq s$ is used precisely in this last Hölder step; equality $p = s$ is allowed.

Let v_j be the full causal potential of g_j . Stop its reversed-time diffusion at exit from $B_{R_j}(x_0)$ or at elapsed time $4r_j^2$. The stopped source integral is $z_j(x_0, t_0)$. At an exit the remaining g_j potential is at most V , because $0 \leq g_j \leq g$; at the strip bottom no source remains. For the trace-one martingale,

$$\mathbb{E}|X_h - x_0|^2 = 2h.$$

The maximal inequality for the nonnegative submartingale $|X_h - x_0|^2$ therefore gives an exit probability at most $8r_j^2/R_j^2$. Hence

$$v_j(x_0, t_0) \leq z_j(x_0, t_0) + 8K_1^{-2} 2^{-2\beta j} V. \quad (33)$$

This is a single stopped decomposition, not an assumed renewal series. The growing relative radius makes its errors summable over all j .

Sum (32)–(33), use $\|g_j\|_{L^p} \leq \|g\|_{L^p}$, and pass $M \rightarrow \infty$ using (31). The source series converges because $\alpha - 2\beta > 0$, and the exit series is at most $V/2$. Thus

$$v(x_0, t_0) \leq C \|g\|_{L^p} + V/2.$$

Taking the supremum over the forcing ball gives $V \leq 2C \|g\|_{L^p}$. For signed smooth g , positivity bounds its potential by that of $|g|$. Smooth nonnegative majorants of $|g|$, supported in a fixed slightly enlarged ball if necessary, converge uniformly and in L^p . Bounded-source comparison lets their potentials pass to the limit. The harmless fixed enlargement changes only structural constants. This proves (30). $\square$

Fix any $p > \max\{p_c, n\Lambda\}$. In dimension two use the ball inverse at $s = 2$. In higher dimensions choose

$$\max\{2, n\Lambda\} < r < \min\{n, p\},$$

which is possible because $n\Lambda < n$, $p > n\Lambda$, and $p > p_c > 2$. Proposition 5.1 supplies the ball inverse at $s = r$; Proposition 7.1 therefore gives the smooth maximum estimate at p . Fixing, for example, the midpoint of this interval makes all constants depend only on n, δ, p .

8 Transfer to the exact measurable-coefficient class

Let A, u satisfy the target assumptions, and let $p > \max\{p_c, n\Lambda\}$. Extend A by I/n outside Q_1 and convolve with nonnegative smooth space-time mollifiers. The resulting globally smooth matrices A_j have bounded derivatives, satisfy (1) with the same δ , and converge to A almost everywhere. Symmetry, trace one, and the lower ellipticity bound are preserved by convexity.

Choose $\eta_m \downarrow 0$ and set

$$D_m = B_{1-\eta_m} \times (\eta_m, 1 - \eta_m).$$

On each fixed closed cylinder the original u_t, D^2u are bounded and continuous. Consequently

$$f_j := u_t - A_j : D^2u \longrightarrow f := \mathcal{P}_A u \quad \text{in both } L^p(D_m) \text{ and } L^{n+1}(D_m)$$

by dominated convergence. In particular, this step never assumes integrability of the original derivatives on all of Q_1 . Choose $g_j \in C_c^\infty(D_m)$ with $\|f_j - g_j\|_{L^p(D_m)} + \|f_j - g_j\|_{L^{n+1}(D_m)} \rightarrow 0$. Let v_j be its whole-space causal potential for A_j on the strip $(\eta_m, 1 - \eta_m)$, extended by zero before the initial time. The smooth maximum estimate gives, uniformly in m, j ,

$$\|v_j\|_\infty \leq C(n, \delta, p) \|g_j\|_{L^p(D_m)}.$$

The unit strip version applies after extending the source by zero; its support remains in $B_1 \times (0, 1)$.

Apply (27) to both signs of $u - v_j$ on D_m . This is legitimate for its stated local $C^{2,1}$ regularity by the local mollification extension of ABP. Set $b_m = \sup_{\partial_{\text{par}} D_m} |u|$. The artificial boundary values of $u - v_j$ have magnitude at most $b_m + \|v_j\|_\infty$, so

$$\|u\|_{L^\infty(D_m)} \leq b_m + 2C \|g_j\|_{L^p(D_m)} + C_{\text{ABP}} \|f_j - g_j\|_{L^{n+1}(D_m)}.$$

Let $j \rightarrow \infty$, then $m \rightarrow \infty$. Uniform continuity of u on $\overline{Q_1}$ and its zero initial and lateral values imply $b_m \rightarrow 0$: compare the artificial initial face with time zero, and the artificial side with its radial projection onto ∂B_1 . Continuity also recovers any maximum at time one. This proves (2) in exactly the original class and finishes the upper-bound part of Theorem 1.1. The argument transfers an estimate; it does not assert a new measurable-coefficient classical existence theory.

9 Sharpness, including both endpoints

Lemma 9.1 (Heat scaling and its endpoint). *Every $1 \leq p \leq (n+2)/2$ is excluded, with $A_j = I/n$ and $u_j \in C_c^\infty(Q_1)$ normalized to have maximum one.*

Proof. Choose a smooth $0 \leq \psi \leq 1$ on $\mathbb{R}^n \times \mathbb{R}$, supported in $\{|y| < 1, |t| < 1\}$ and equal to one on $\{|y| \leq 1/2, |t| \leq 1/4\}$. Put $t_0 = 1/2$, $r_j = 4^{-j}$ and

$$\psi_j(x, t) = \psi(x/r_j, (t - t_0)/r_j^2), \quad L = \partial_t - \Delta/n.$$

Each cutoff is compactly supported in Q_1 and equals one at $(0, t_0)$. The support of the next cutoff lies strictly inside the preceding plateau, so the supports of $L\psi_j$ are pairwise disjoint. Scaling gives

$$\|L\psi_j\|_{L^p}^p = r_j^{n+2-2p} \|L\psi\|_{L^p}^p.$$

For $p < p_c$ take $u_j = \psi_j$. At $p = p_c$ take $u_M = M^{-1} \sum_{j=1}^M \psi_j$. Its maximum is one, and disjointness gives

$$\|Lu_M\|_{L^{p_c}}^{p_c} = M^{1-p_c} \|L\psi\|_{L^{p_c}}^{p_c} \longrightarrow 0.$$

Since $I/n \geq \delta I$ and has trace one, these are examples in the exact coefficient class, with all prescribed boundary values zero. $\square$

Lemma 9.2 (Radial obstruction and its endpoint). *Every $1 \leq p \leq n\Lambda$ is excluded by smooth coefficient fields and classical functions satisfying exactly the required parabolic boundary condition, again normalized to have maximum one.*

Proof. Write $d = 1/\Lambda$, so $1 < d < n$. Fix $0 < \eta < 1/16$ and choose a smooth radial function a_η , with values in $[1/n, \Lambda]$, equal to $1/n$ on $[0, \eta]$ and to Λ on $[2\eta, 1]$. Set

$$b_\eta = \frac{1 - a_\eta}{n - 1}, \quad A_\eta(x) = a_\eta(r)e_r \otimes e_r + b_\eta(r)(I - e_r \otimes e_r), \quad r = |x| > 0,$$

and $A_\eta(0) = I/n$. It is smooth at zero because it is constant there; it is symmetric, trace one, and every eigenvalue is at least δ . On radial functions its generator is

$$L_\eta v = a_\eta v'' + \frac{1 - a_\eta}{r} v'.$$

Define positive m_η and μ_η by

$$\frac{m'_\eta}{m_\eta} = \frac{1 - a_\eta}{a_\eta r}, \quad m_\eta(r) = r^{d-1} \quad (r \geq 2\eta), \quad \mu_\eta = \frac{m_\eta}{a_\eta}.$$

This normalization is consistent because $(1 - \Lambda)/\Lambda = d - 1$. Then

$$L_\eta v = \frac{1}{\mu_\eta} (m_\eta v')'. \quad (34)$$

Since $(1 - a_\eta)/a_\eta \geq d - 1$, integration towards zero yields

$$m_\eta(r) \leq r^{d-1}, \quad \mu_\eta(r) \leq nr^{d-1}, \quad W_\eta := \int_0^1 \mu_\eta dr \leq n/d.$$

For $r \geq 2\eta$, the exact formula is $\mu_\eta = r^{d-1}/\Lambda$. Near zero m_η is a constant multiple of r^{n-1} .

Take a nonzero smooth nonnegative radial f_η , supported in $2\eta < r < 1/2$, and solve

$$(v_\eta)_t - L_\eta v_\eta = f_\eta, \quad v_\eta = 0 \text{ on } \partial_{\text{par}} Q_1.$$

For each fixed η this is a smooth uniformly elliptic Dirichlet problem. Its source vanishes near the lateral boundary, so the initial/lateral corner compatibility conditions hold. Classical existence gives $v_\eta \in C(\overline{Q_1}) \cap C^{2,1}(Q_1)$, and the maximum principle gives $v_\eta \geq 0$. Uniqueness makes it radial. No derivative estimate uniform in η is used.

With $\phi(r) = 1 - r^2$, put

$$U_\eta(t) = \int_0^1 v_\eta \phi \mu_\eta dr, \quad V_\eta(t) = \int_0^1 v_\eta \mu_\eta dr, \quad F_\eta = \int_0^1 f_\eta \phi \mu_\eta dr.$$

Trace one gives $L_\eta \phi = -2$. The self-adjoint identity from (34) yields

$$U'_\eta = F_\eta - 2V_\eta.$$

Its boundary term is $[m_\eta(\phi v'_\eta - \phi' v_\eta)]_0^1$. At one both functions vanish; at zero smooth radially gives $v'_\eta = O(r)$, $\phi' = O(r)$ and $m_\eta = O(r^{n-1})$, so the term vanishes. The time-zero limit is justified by continuity and $W_\eta < \infty$. If $M_\eta = \max_{\overline{Q_1}} v_\eta$, integration from zero to one gives

$$F_\eta = U_\eta(1) + 2 \int_0^1 V_\eta(t) dt \leq 3W_\eta M_\eta \leq (3n/d)M_\eta.$$

Thus $M_\eta > 0$, and $u_\eta = v_\eta/M_\eta$ satisfies

$$\max |u_\eta| = 1, \quad \|\mathcal{P}_{A_\eta} u_\eta\|_{L^p(Q_1)} \leq \frac{3n}{d} \frac{\|f_\eta\|_{L^p(B_1)}}{F_\eta}. \quad (35)$$

For $p < n/d$, choose a fixed nonzero nonnegative $\chi \in C_c^\infty((3, 4))$ and set $f_\eta(r) = \chi(r/\eta)$. Then

$$F_\eta \geq c\eta^d, \quad \|f_\eta\|_{L^p(B_1)} = C_p \eta^{n/p}.$$

The ratio in (35) tends to zero. At $p = n/d$, take smooth cutoffs so that

$$f_\eta(r) = r^{-d} \theta(r/\eta) \zeta(r),$$

where $\theta = 0$ on $[0, 3]$, $\theta = 1$ on $[4, \infty)$, $\zeta = 1$ on $[0, 1/4]$, and $\zeta = 0$ on $[1/2, \infty)$. This is smooth, being zero near the origin. The exact speed measure gives

$$F_\eta \geq c \log(1/\eta), \quad \|f_\eta\|_{L^{n/d}(B_1)}^{n/d} \leq C \log(1/\eta).$$

Hence the normalized residual is at most $C[\log(1/\eta)]^{d/n-1} \rightarrow 0$, since $d < n$. $\square$

The two lemmas exclude the union of $[1, p_c]$ and $[1, n\Lambda]$, namely $[1, \max\{p_c, n\Lambda\}]$, including its endpoint. The upper proof established every larger finite exponent in exactly the prescribed class. This completes the proof of Theorem 1.1.

Analytic dependencies and verification

The only nonlinear elliptic inputs are the strictly positive Dirichlet solvability theorem and the stated Hessian Sobolev inequality. The new matrix inequality, the Mellin cancellation, convex source splitting, and all boundary flux signs have complete proofs above. Other standard tools are classical smooth linear parabolic solvability and positivity, the maximum principle, the area formula, and the stopped diffusion representation for smooth bounded uniformly elliptic coefficients. The last is used solely in the expanding-annuli localization, with its exit probability explicitly estimated by the trace-one second-moment identity.

In particular the proof does not invoke a coefficient-uniform classical existence theorem for measurable coefficients, an unidentified critical Green-function integrability theorem, a fractional Hessian operator, or an unproved interpolation equivalence. Measurable coefficients and the exact locally classical test class enter only through the explicit interior approximation and ABP residual comparison.

This is an ordinary written mathematical proof, checked through separate reconstructions of its algebraic and analytic interfaces, including independent language-model audits. It is not a formally certified proof.

References

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